HipMRI Segmentation: Standard 3D U-Net and Residual Improved U-Net v3

COMP3710 Final Assessment | Student ID: s4906923 | Final report updated 22 October 2026

1. Abstract

This study compared ten completed training runs on a patient-disjoint HipMRI cohort: 26 Train, 6 Validation and 6 Test patients. Standard 3D U-Net and residual Improved v3 were selected for the final runnable package; every reported model uses its own best measured Validation checkpoint. v3 raised patient-equal Test small-organ Dice from 0.7851 to 0.8145 and five-foreground Dice from 0.8370 to 0.8631. The prostate gain was smaller, from 0.7433 to 0.7485. Median prostate HD95 improved from 7.98 to 6.65 mm, but the worst HD95 increased from 19.23 to 115.37 mm. Existing Test predictions also show that both 3D configurations improve end-region overlap and slice continuity over the supplementary 2D configuration, while v3 does not improve those measures over Standard. Thus, the experiments support useful average segmentation gains, but do not establish consistently better local contours or reliable automated contouring.

2. Introduction

Research focus: the spatial context versus clinical boundary utility dilemma. High whole-organ Dice can conceal errors at the prostate base and apex, discontinuities between slices, and predictions extending into nearby bladder or rectum. The practical question is whether true 3D continuity and the Improved model's multiscale outputs reduce these local failures compared with a standard 2D baseline, or mainly improve easy central slices while increasing computation.

This report investigates three linked questions. **RQ1:** Where do the tested 3D configurations improve prostate end-region overlap and slice continuity relative to the supplementary 2D configuration? **RQ2:** Does Improved v3 improve local boundaries and adjacent-organ separation beyond Standard 3D, including extreme errors that average Dice can hide? **RQ3:** Do those anatomical gains justify the measured memory and runtime budget in an operational contour-drafting workflow?

The study concerns HipMRI rather than OASIS or CAN architectures. It measures segmentation overlap, physical surface errors, slice-transition proxies and adjacent-organ confusion. It does not measure radiation dose, rectal-wall or bladder-mucosa delineation, treatment toxicity or cancer outcomes. Therefore, the clinical motivation defines which errors matter; it does not establish clinical benefit or safety from these experiments.

The existing Test cohort was inspected during the research process, so these findings are descriptive rather than a fresh blind assessment. The final recommendation is to retain v3 as the leading candidate under the original Validation rule, keep Standard as the reference model, and require human review of organ boundaries. No model is claimed to be best for every patient or metric.

Report organisation. Section 3 documents the pretraining data checks, the two final architectures, the training protocol and the development of all ten model configurations. Section 4 presents the measured comparisons and five image-based failure analyses; Section 5 gives the engineering recommendation. The one-page feasibility review is retained in Appendix C, within this README as required by the coursework.

3. Research methods

3.1 Pretraining data investigation, preprocessing and spatial integrity

Pretraining data investigation and acceptance decision. Training began only after checking the source pairs, the actual preprocessing path, the delivered samples and basic learning behaviour. The initial audit decoded Train arrays only; Validation/Test filenames were inventoried to check the split, without using their label distributions to design the recipe.

Check before formal cohort training	Recorded evidence and decision
Cohort and mechanical integrity	All 143 Train MRI/mask pairs from 26 patients were audited; no inventory/case errors, non-finite values or illegal labels were found. All six labels occurred in every Train volume; no identical-file hash groups were found within Train.
Geometry and gross alignment	Every MRI/mask affine matched exactly. All Train MRI used LPS orientation and millimetres; 142 volumes were 256x256x128 and one was 256x256x144. Two axial overlays per volume showed no obvious gross displacement. This screen did not certify every local contour.
Documented metadata warning	All 143 mask headers left spatial units unspecified. Matching grids, affines and spacings supported using the paired MRI geometry, with this limitation retained rather than silently correcting the source.
Distribution and supervision	Train-only counts showed Background 59.99%, Body 35.75%, Bones 3.41%, Bladder 0.60%, Rectum 0.15% and Prostate 0.11%. Visit counts varied from 1 to 8 per patient. This motivated patient-balanced sampling and Train-derived CE weights, rather than treating correlated visits as independent patients.
Actual preprocessing	The factor-two pipeline processed 143/143 Train cases without errors; all small organs survived. Prostate physical label volume after/before resampling ranged from 0.9383 to 1.0423. Normalised nonzero MRI values had mean approximately 0 and standard deviation approximately 1. Organ survival did not prove thin-boundary preservation.
Delivered samples and learning checks	Patient-balanced CPU sampling/tensor checks preceded a paired GPU optimiser smoke and full-volume capacity probe. In a controlled two-patch, 500-update check, unweighted CE left Standard predicting no small-organ voxels; Train-derived weights enabled fitting those same organs. This justified a weighted learning protocol, not a claim about unseen-patient accuracy.

Go/no-go interpretation. Zero unresolved mechanical errors, matching paired geometry, preserved labels through the actual transform, finite model inputs, verified sampling and a working optimiser/inference path supported proceeding to controlled training. They established usable engineering inputs, not expert-certified annotations or clinical suitability. The unspecified mask units and downsampling's potential loss of millimetre-scale detail remained explicit limitations. The numerical summaries and original audit hashes are retained under `pretraining_review` in `evidence/provenance.json`.

Split	Patients	Volumes	Purpose
Train	26	143	Optimisation and Train-derived weights
Validation	6	38	Checkpoint selection and diagnosis
Existing Test	6	30	Descriptive fixed-checkpoint evaluation

`config/patient_split.json` records the fixed patient lists. Visits from one patient never cross splits. Files are paired by patient/week. `dataset.py` checks matching 3D shape, affine and voxel spacing, finite arrays and integer labels 0-5. MRI must declare millimetres; matching masks with unknown spatial units are accepted as a documented source-metadata limitation. Metadata agreement is not expert confirmation of anatomy.

MRI and mask are jointly downsampled by (2,2,2). Output centre j samples source centre `(j+0.5)*2-0.5`; the affine uses the same transform. MRI interpolation is linear, mask nearest-neighbour; lost labels trigger failure. Image-only nonzero Z-score preserves outside zeros. The ground-truth mask (GT) never defines the region used for prediction.

Arrays use X/Y/Z voxel order until conversion to channel/depth/height/width (CDHW) model input. High-end zero padding makes dimensions divisible by 16 and is removed before metrics/export. Evaluation is on the downsampled physical grid, not original resolution. Cross-entropy (CE) takes integer labels; Dice converts them to six-channel one-hot internally; `one_hot()` is also available.

Keep raw files outside Git:

```
HipMRI_Study_open/
  semantic_MRs/PatientID_WeekN_LFOV.nii.gz
  semantic_labels_only/PatientID_WeekN_SEMANTIC.nii.gz
```

3.2 Final model architectures: Standard 3D and Improved v3

Why these architectures? Standard supplies an implemented volumetric encoder-decoder reference. The residual Improved family adapts Isensee-style context modules and multiscale supervision to the coursework Hard route. The compact starting width of eight channels makes the downsampled experiments feasible; it is a project choice rather than a full-size reproduction of a cited paper. The following specification describes the networks without requiring the reader to inspect their implementation.

Property	Standard 3D U-Net	Improved 3D U-Net v3
Input and output	One MRI channel; six class logits	One MRI channel; six combined class logits
Feature hierarchy	Five resolutions, channels 8,16,32,64,128; four downsampling steps	Same five resolutions and channel widths
Encoder stage	Two 3x3x3 convolutions, each followed by GN1 and ReLU	3x3x3 stem; a residual context block at every resolution
Downsampling	2x2x2 max pooling, stride 2	3x3x3 convolution, stride 2, padding 1; channels double
Decoder step	2x2x2 transposed convolution, stride 2; concatenate matching encoder features; two-convolution block	Nearest-neighbour upsampling to the skip grid, 3x3x3 channel reduction and LeakyReLU; concatenate matching context features; localisation block
Localisation / block details	3x3x3 convolutions use padding 1 and no bias	Localisation: 3x3x3 convolution on 2C channels, GN1 and LeakyReLU, then 1x1x1 reduction to C
Normalisation and activation	GN(1,C), epsilon 1e-5, learned scale/offset; ReLU	Fourteen GN(1,C) sites, epsilon 1e-5, learned scale/offset; LeakyReLU slope 0.01
Residual and regularisation	No residual blocks or dropout	Five identity residual blocks; Dropout3d probability 0.3 between each block's two convolutions, active during training
Segmentation heads	One 1x1x1 convolution at full resolution	1x1x1 heads at full, half and quarter resolution; resize the latter two and sum the logits
Trainable parameters	1,412,550	2,366,946

For a training patch, both encoders follow this spatial/channel schedule. Sizes are X/Y/Z voxels; the implementation receives Bx1x32x64x64 after conversion to depth/height/width order. B is batch size.

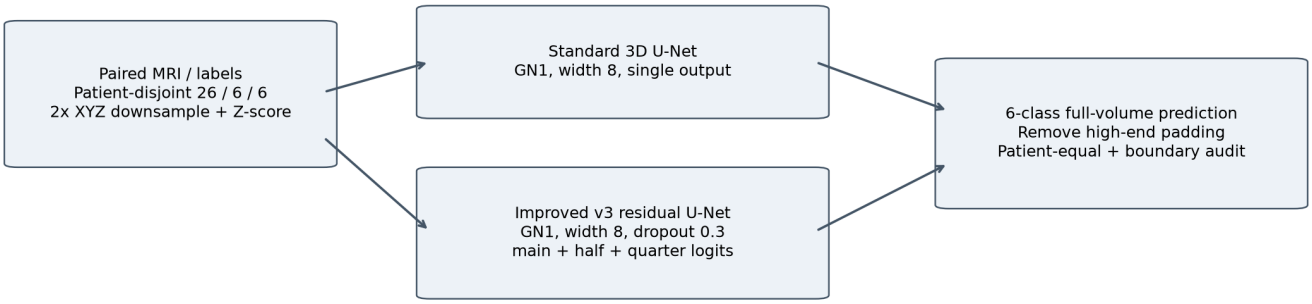
Encoder level	Channels	Patch feature size X/Y/Z
Full resolution	8	64x64x32
Half resolution	16	32x32x16
Quarter resolution	32	16x16x8
One eighth	64	8x8x4
Bottleneck, one sixteenth	128	4x4x2

The decoder reverses this hierarchy, using four feature-concatenation skips, and ends with Bx6x32x64x64 logits for a training patch. Evaluation accepts a larger padded whole volume with the same channel hierarchy; this is not a fixed patch-size inference model.

The v3 residual block. For feature tensor x with C channels, the branch applies GN1 -> LeakyReLU -> 3x3x3 convolution -> Dropout3d -> GN1 -> LeakyReLU -> 3x3x3 convolution, then adds the original x. The spatial convolutions use stride 1 and padding 1. GN1 normalises across channels and spatial positions within each sample, not across the batch; this differs from v1's per-channel InstanceNorm and can interact with input extent.

Output fusion and supervision. The full-resolution head sees eight decoder channels; the half- and quarter-resolution heads see 16 and 32. Auxiliary logits are resized to the main grid by trilinear interpolation with `align_corners=False`. Training and inference both use `combined = main + resized_half + resized_quarter`; one loss is applied to this

sum, so all three heads receive gradients. This is additive multiscale supervision, not the independent head losses of v2. Class order is Background, Body, Bones, Bladder, Rectum, Prostate. Softmax supplies probabilities for the Dice loss; inference uses the largest combined logit per voxel.



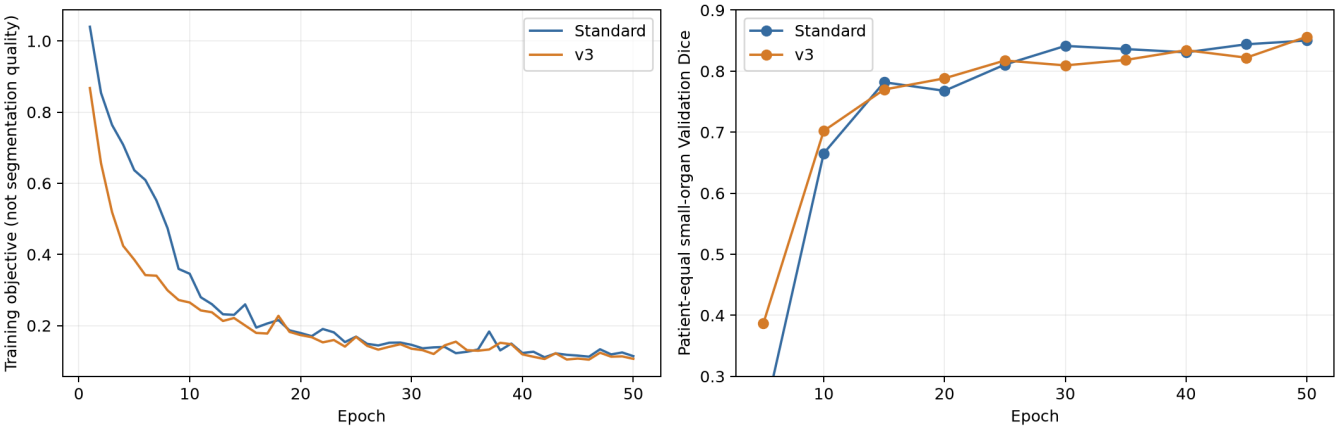
Data and model pipeline

3.3 Training objective, sampling and checkpoint selection

Both models use random seed 371017 and the Adam optimiser, with a fixed learning rate of 0.001 and no weight decay. One shuffled/cycled visit per Train patient supplies ten uniform 64x64x32 XYZ patches per epoch: 260 patches, batch size 2, 130 updates; 50 epochs yield 6,500 updates. The submission preserves the deterministic visit/crop/batch schedule. Exact GPU rerun weights are not guaranteed across kernels, hardware and library versions.

Loss is `0.5 *weighted_cross_entropy* + 0.5 foreground_soft_Dice_loss`. Background remains in CE but is excluded from Dice. CE weights derived from Train data are `[1.453871, 1, 2.617199, 4.498586, 8.893215, 9.699950]` in label order. Giving small organs more weight can improve recall at the cost of precision. Lower training loss is not proof of better segmentation.

Validation runs every five epochs. Pool confusion matrices across visits within each patient; compute Bladder/Rectum/Prostate Dice, average within patient, then average six patients equally. Choose the highest score at measured epochs 5,10,...,50, with earlier ties. The intermediate epochs were not all evaluated. Both selected checkpoints happen to be epoch 50.



Historical training and checkpoint selection

3.4 Model development: observations, hypotheses and controlled changes

Development logic. The study completed ten formal 50-epoch runs: Standard, supplementary 2D, v1, v2, v3, A0, A1 and v5-C/L/S. Tiny-batch checks, inference probes and sampler audits were diagnostics, not additional successful cohort models. The sequence used Train exposure records and Validation predictions to decide what to test next. Recorded configuration changes below describe intended mechanisms; they are not by themselves proof of the cause of an error. The complete numerical comparisons are in Section 4. Standard, v1 and the supplementary 2D run were extended from the initial 40-epoch budget to 50; the final comparison uses the same 50-epoch total budget for every run. Earlier v2/v3 mechanism plans also specified fixed-epoch checks. Those historical checks are distinct from the final benchmark, which consistently reports each model's best measured Validation checkpoint.

Starting comparison and diagnostic loop. Standard and the first Improved network, subsequently called v1, were given the same patient-balanced 3D crops, loss, split and optimisation recipe. The residual context, InstanceNorm, dropout and additive output heads in v1 were intended to add contextual representation and regularisation. After training, review moved from loss curves to per-class Validation confusion counts, precision/recall, physical Body surfaces and MRI overlays. Reproducing the fixed predictions before calculating new measurements prevented a change of checkpoint or preprocessing from masquerading as a diagnostic finding.

The striking v1 Body error prompted competing explanations: insufficient boundary/background exposure; coupled output heads; imbalance weights; and normalisation or full-volume context. Train geometry/exposure audits tested what the sampler supplied. Fixed-weight Validation probes compared full-volume and sliding-window inputs and inspected the individual output heads. Neither inference-context changes nor removal of auxiliary outputs was treated as a retrained architecture comparison. Small numerical differences caused a head-replay control to fail; that control was not used as causal proof. These limits motivated bounded training interventions rather than assuming one established root cause.

Run / parent	Design decision and question tested	Comparison logic
Standard / initial reference	Basic 3D U-Net, GN1, one head; establish a volumetric baseline under the verified pipeline	Same 3D split, patch schedule, objective and update budget as v1
v1 / initial Improved	Residual context, InstanceNorm, dropout 0.3 and summed multiscale logits; test the contextual Hard architecture	Several components differ from Standard, so the aggregate contrast cannot identify one causal component
2D / supplementary branch	Width-16 slice U-Net; four uniform plus two crops per small organ per patient, with its own Train-derived weights	Slice-only context and lower resource use provide an engineering reference; different width/sampling/weights prevent a pure dimensionality ablation
v2 / v1	Train main, half and quarter heads independently: `(Lmain + 0.5 Lhalf + 0.25 Lquarter) / 1.75`; use the main head at inference	Tests whether individually useful heads avoid reliance on summed predictions; keeps v1's backbone and InstanceNorm. Training supervision and inference output change together
v3 / v1	Replace all 14 InstanceNorm sites with GN1; retain v1's residual blocks, dropout, additive heads and loss	Isolates the declared normalisation change relative to the historical v1 recipe; does not inherit v2's independent losses
A0 / v3	AdamW, decay 1e-5 excluding bias/normalisation, cosine learning rate from 0.001 towards 1e-5 over 6,500 updates	Tests a conventional optimisation/regularisation package; the combined changes cannot isolate decay from optimiser or schedule
A1 / A0	With probability 0.3, apply paired physical-axis rotations up to 10 degrees and isotropic scaling 0.9-1.1; linear MRI and nearest-neighbour mask interpolation	A1-A0 tests this specific spatial augmentation under the AdamW/cosine recipe. A Train-only 26-patient geometry/exposure audit and real previews preceded training
v5-C / v3	Concurrent run using the original v3 settings and nominal seed	Provides a contemporary control for L/S and reveals same-recipe run variation; this is not an independent-seed replication
v5-L / C	Compress only the three small-organ CE weights by square root: 4.498586/8.893215/9.699950 become 2.120987/2.982149/3.114474	Tests a less aggressive imbalance penalty while retaining Adam, constant learning rate, architecture and sampling
v5-S / C	Retain Adam and zero decay; change only the learning-rate rule to cosine over 6,500 updates, towards 1e-5	Separates the schedule question from A0's additional optimiser/decay changes

How each redesign followed the evidence. v2 was a targeted response to the head-coupling hypothesis, not a cosmetic extension of v1. Its independently supervised design did not resolve the overall Validation trade-off, so v3 returned to v1 and tested the alternative normalisation hypothesis. v3's improvement justified retaining it as a reference for further research, while residual boundary errors motivated the A0/A1 regularisation and spatial-exposure experiment. A1 was accepted for training only after paired-transform checks; passing those checks supported mechanical validity, not effectiveness. The absence of an overall gain from that package then motivated C/L/S, which separated a smaller class-weight intervention and the learning-rate schedule from other changes. The Train edge-sampling candidate was audited but never trained as a formal segmentation run; it is not counted among the ten models.

Why the final report centres on Standard and v3. The original selection endpoint remained patient-equal Validation Dice for Bladder, Rectum and Prostate, measured every five epochs. Each run first contributes its best measured checkpoint; comparing those checkpoints places v3 first under this rule. The selection advantage over Standard and C is small, not evidence of universal superiority. Body/surface diagnostics also show that v3 mitigates the severe failure that motivated revising v1, while keeping the residual/multiscale architecture. No later branch supplied a better overall Validation candidate. Standard remains the operational reference because its simpler architecture and boundary behaviour remain informative. C's

higher existing Test prostate score does not justify changing the rule after seeing Test; v3 is the selected research candidate, not the winner of every metric or a clinically approved model. Section 4 quantifies those gains and regressions; Section 5 weighs their practical importance.

3.5 Evaluation definitions and limitations

All models use identical patient splits and evaluation grids. 3D inference processes the complete downsampled volume once, not sliding windows. The supplementary 2D model processes slices and reconstructs a volume. No GT-dependent post-processing is used. Taking the largest class logit gives the same class as applying softmax first.

TP, FP and FN denote true positives, false positives and false negatives. Dice=`2TP/(2TP+FP+FN)`, intersection over union (IoU)=`TP/(TP+FP+FN)`, precision=`TP/(TP+FP)`, recall=`TP/(TP+FN)`. If a class is absent from both prediction and GT, its overlap is recorded as unavailable rather than perfect. Primary overlap tables pool visits within patients before averaging patients. The foreground mean gives equal weight to labels 1-5; Background remains in the six-class table. The supplementary post-hoc 50% prostate-weighted score is not a selection rule or official course criterion.

The 95th-percentile Hausdorff distance (HD95) and average symmetric surface distance (ASSD) use concatenated directed distances between six-connected mask surfaces, in physical millimetres. HD95 is the 95th percentile; ASSD here is their combined mean. Empty-mask statuses are explicit and null, never zero distance. All five foreground masks were present in all 680 fixed-audit case/model entries. Surface medians/maxima below summarise volumes, unlike patient-equal overlap. Other libraries may use different surface weighting conventions. The new diagnostic refuses sheared grids where spacing-only Euclidean distances would be invalid.

Test has already been repeatedly inspected. Checkpoints were chosen using Validation. Because Test has already been examined, it is not a fresh blind holdout and should not guide further tuning. Six patients, not 30 correlated visits, are the independent Test units. No external cohort or independent multi-seed replication was completed.

4. Results and model comparisons

4.1 Best Validation checkpoints

Every model in the table completed 50 training epochs. Only Standard/v3 runtime implementations are retained here; the other experiments are retained as comparisons, including changes that did not improve performance.

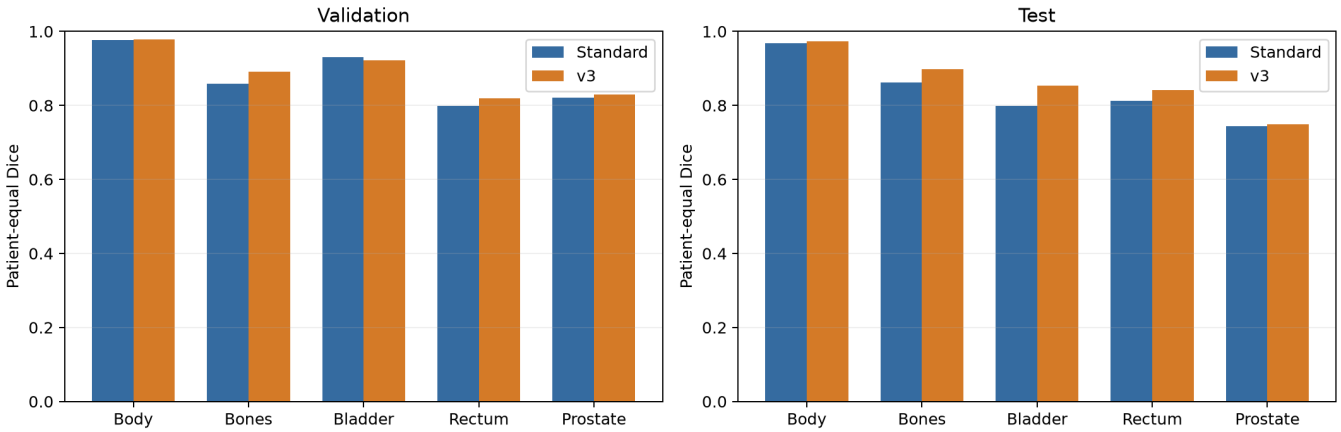
Model	Best epoch	Validation small organs	Test small organs	Test foreground five	Test Prostate
2D	50	0.7757	0.7082	0.7883	0.6611
Standard	50	0.8502	0.7851	0.8370	0.7433
v1	50	0.8420	0.8078	0.8502	0.7520
v2	45	0.8362	0.7959	0.8422	0.7478
v3	50	0.8564	0.8145	0.8631	0.7485
A0	45	0.8450	0.7836	0.8400	0.7318
A1	45	0.8298	0.7801	0.8362	0.7273
v5-C	50	0.8542	0.8040	0.8557	0.7633
v5-L	45	0.8426	0.7858	0.8388	0.7361
v5-S	45	0.8465	0.7788	0.8350	0.7281

A1 augmentation result (design in Section 3.4). The best Validation small-organ Dice decreased from 0.8450 to 0.8298, and existing Test Dice from 0.7836 to 0.7801. Distant Body false positives decreased, but this did not yield an overall segmentation gain. This result concerns one tested augmentation scheme and training budget; it does not show that data augmentation is generally ineffective.

v3's Test small-organ gain over Standard is +0.0294, but only 3/6 patients improve; excluding influential M015 changes the contrast to approximately -0.0050. Validation improves for 5/6 patients. These descriptive sensitivities do not establish population superiority. C's higher Test prostate Dice is reported without switching the primary selection rule after seeing Test.

4.2 Test results by class

Class	Std Dice	v3 Dice	Std IoU	v3 IoU	Precision Std / v3	Recall Std / v3
Background	0.9904	0.9885	0.9814	0.9778	0.9952 / 0.9961	0.9858 / 0.9814
Body	0.9677	0.9732	0.9384	0.9484	0.9748 / 0.9664	0.9610 / 0.9807
Bones	0.8619	0.8988	0.7578	0.8168	0.7868 / 0.8793	0.9543 / 0.9205
Bladder	0.7993	0.8539	0.7455	0.7768	0.7871 / 0.8807	0.8229 / 0.8598
Rectum	0.8128	0.8410	0.6918	0.7280	0.7898 / 0.8086	0.8414 / 0.8784
Prostate	0.7433	0.7485	0.6081	0.6193	0.7189 / 0.6983	0.8415 / 0.8516



Patient-equal foreground Dice

v3 improves all five Test foreground mean Dice values here, but prostate gains only about 0.0052 while precision decreases from 0.7189 to 0.6983. Mean overlap can improve despite more false-positive prostate predictions, and individual cases regress.

4.3 Physical boundary reliability

Class	Median HD95 Std / v3 mm	Worst HD95 Std / v3 mm	Median ASSD Std / v3 mm
Body	4.86 / 4.50	18.72 / 22.23	1.21 / 1.66
Bones	7.46 / 4.58	18.97 / 19.02	2.69 / 1.83
Bladder	3.36 / 3.89	109.49 / 72.36	1.38 / 1.51
Rectum	7.06 / 7.28	38.56 / 109.19	2.46 / 2.09
Prostate	7.98 / 6.65	19.23 / 115.37	3.45 / 2.73

Median prostate HD95 improves 7.98 to 6.65 mm, while the worst value worsens 19.23 to 115.37 mm. Body/Rectum also retain severe tails. The Hard-route average target of 0.7 is exceeded by v3's Test prostate (0.7485), small-organ (0.8145) and foreground (0.8631) means. The specification does not fully define aggregation; one of six Test patients still has prostate Dice below 0.7. Meeting an average target does not establish local contour safety.

4.4 Resources and spatial context

Model	Parameters	Peak inference GiB	Reference latency s
2D	1,942,374	0.0761	0.1395
Standard	1,412,550	0.2460	0.1552
v3	2,366,946	0.3458	0.1452

NVIDIA A100-PCIE-40GB, prepared Validation case K008_Week0, three warm-ups and five repetitions, one resident model. Latency includes transfer, execution, argmax and conversion, excluding decoding/preprocessing/metrics/IO. Peak figures are

allocated inference memory, not training memory or device utilisation. Timing depends on shared load/kernels; parameter count is not runtime. Original per-case times and per-patient counts are in the evidence.

The 2D model uses less GPU memory (VRAM) during inference but its kernels cannot directly see adjacent slices. 3D context can help continuity yet does not ensure correct prostate ends or exclude distant components. The supplementary 2D arm differs in sampling, weights and width; it is not a clean dimension-only ablation and does not claim to meet the separate Easy-route dataset/target. Measurements of prostate end regions, nearby organs at risk (OAR) and changes between adjacent slices remain in `evidence/experiments.json`.

4.5 Prostate ends, slice continuity and adjacent-organ errors

The following measurements bring the spatial research question into the report directly. They are calculated from the existing predictions and measurements already recorded for the 30 Test volumes; no new inference, tuning or checkpoint selection was performed.

Existing Test diagnostic	2D	Standard 3D	Improved v3
Apex mean slice Dice	0.5186	0.7544	0.7523
Central mean slice Dice	0.7125	0.7824	0.7931
Base mean slice Dice	0.3473	0.5097	0.5069
Visits with an internal predicted prostate gap (%)	43.75	0.00	8.33
Adjacent-area transition error (dimensionless)	0.1953	0.0685	0.0720
GT Bladder/Rectum voxels labelled Prostate (%)	0.386	0.485	0.580

End-region values average GT-positive slice Dice within each visit, then visits within each patient, and finally the six patients equally. Base and apex are the physically superior and inferior 20% of GT-positive slices: geometric proxies rather than clinician-defined regions. These slice averages differ from the pooled whole-organ Dice in Section 4.2.

The gap statistic is the patient-equal mean fraction of visits with an empty predicted Z slice between nonempty predicted prostate slices. The GT has no such gaps in this cohort. A lower value is consistent with better slice presence, but a continuous prediction can still be in the wrong location. The area-transition error is the absolute difference between prediction and GT mean normalized adjacent-area changes on GT-positive adjacent slice pairs. It is a descriptive shape-change measure, not a validated anatomical quality score.

The adjacent-organ percentage pools counts within each patient before averaging patients. Its denominator is all GT Bladder/Rectum voxels, and its numerator is those labelled as Prostate. It measures class confusion, not specifically wall invasion, mucosal dose or clinical harm. The small absolute percentages should not obscure local errors near a clinically important interface. Definitions and unrounded values are retained in `evidence/spatial_context_summary.json`.

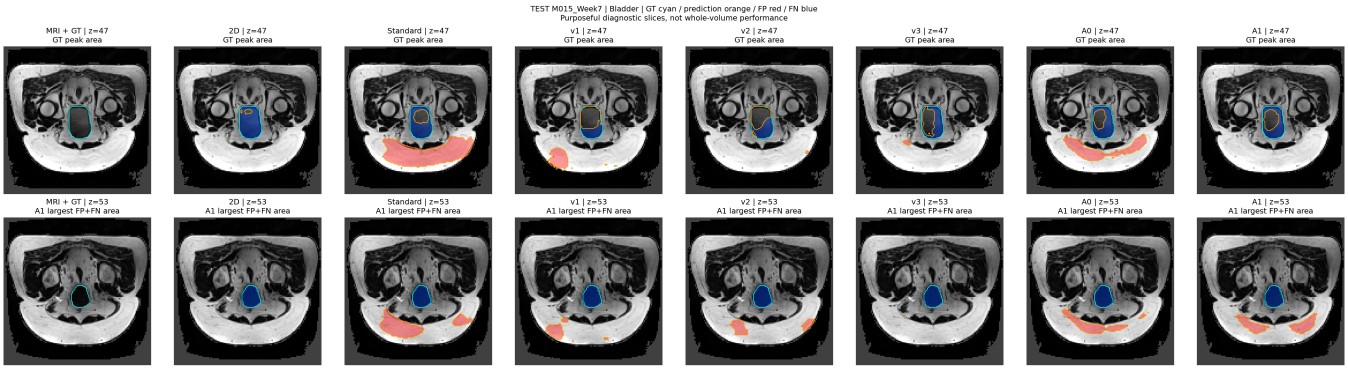
Interpretation. Standard/v3 improve apex and base overlap over this 2D configuration and show fewer internal gaps. However, v3's gains over Standard are concentrated in central slices; its apex and base means are slightly lower, its continuity measures are not better, and its adjacent-organ confusion percentage is higher. These results directly qualify the favourable whole-organ average: more sophisticated 3D processing did not consistently solve the boundary problem.

4.6 Five real-image failure autopsies

These unchanged historical fixed-checkpoint panels are real MRI, not simulated or retouched contours. Cyan=GT, orange=prediction, red=FP, blue=FN. Seven labelled historical model columns are retained for provenance; compare **Standard** and **v3** for the submitted pair. Open the linked PNGs at full resolution to inspect small structures.

Ordinary panels show the GT-largest-area slice and the slice with largest A1 FP+FN area, shared across models. The Body panel uses A1 maximum-error planes. This purposive selection is not a random sample or a whole-volume score. Prostate base/apex uses physical superior/inferior ordering as a geometric proxy, not expert end-region annotation.

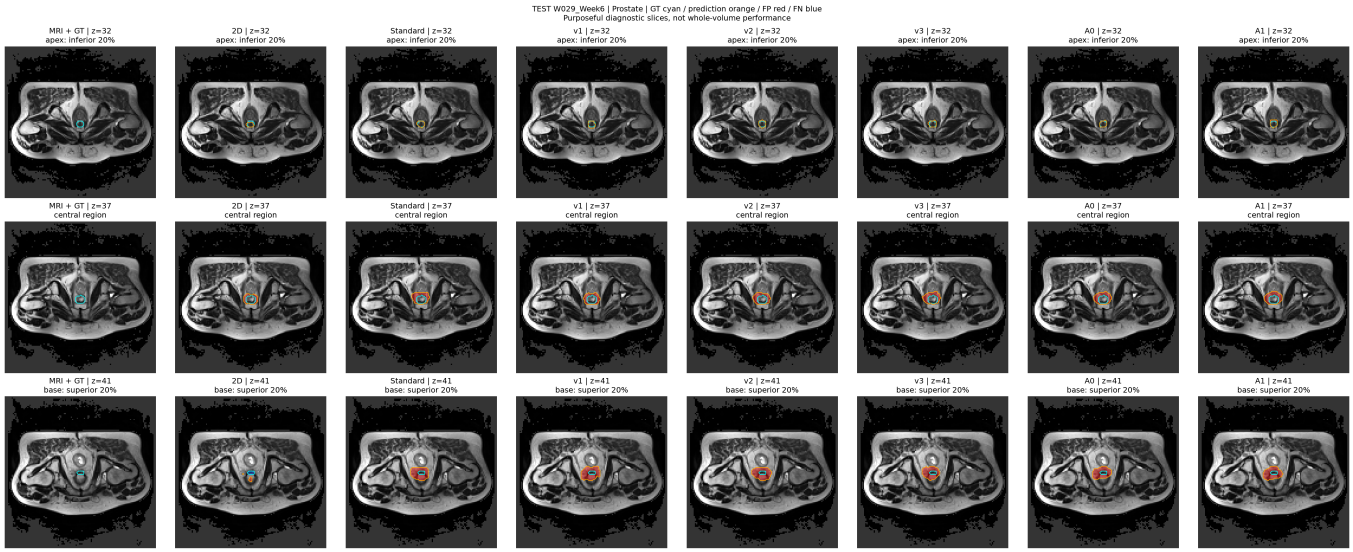
4.6.1 M015_Week7, Test: bladder localisation



M015 bladder errors

Standard includes distant non-bladder tissue and misses much of the bladder; v3 reduces distant predictions but still undersegments the target. Whole-volume bladder Dice is 0.0676 Standard versus 0.5224 v3; HD95 is 109.49 versus 58.84 mm. This is improvement within a poor case, not a successful contour. Patient-specific anatomy/intensity is a plausible trigger, not an established cause. The patient's influence requires sensitivity analysis rather than assuming the Test mean represents universal gain.

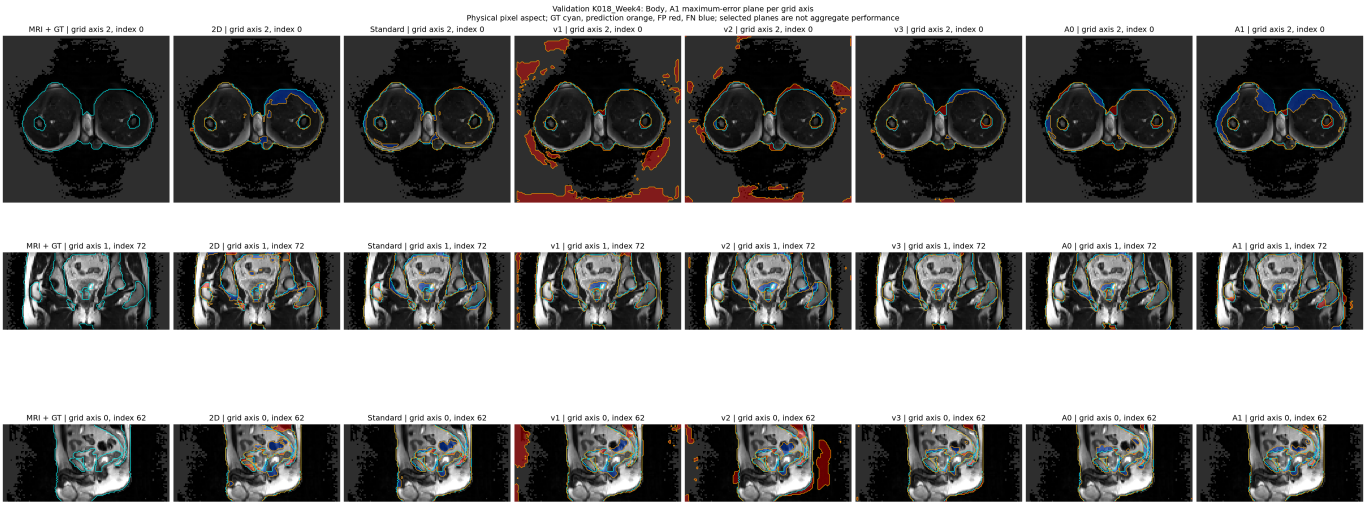
4.6.2 W029_Week6, Test: prostate ends and adjacent tissue



W029 prostate ends

Both models have poor base overlap and extend beyond GT. Whole-volume prostate Dice is 0.4623 Standard versus 0.4377 v3; base-region mean slice Dice is 0.2111 versus 0.2158. Slight end-region improvement does not imply whole-organ improvement. HD95 increases 19.23 to 20.59 mm. Shrinking end cross-sections and neighbouring structures make precision difficult; no dose impact or expert clinical adjudication was measured.

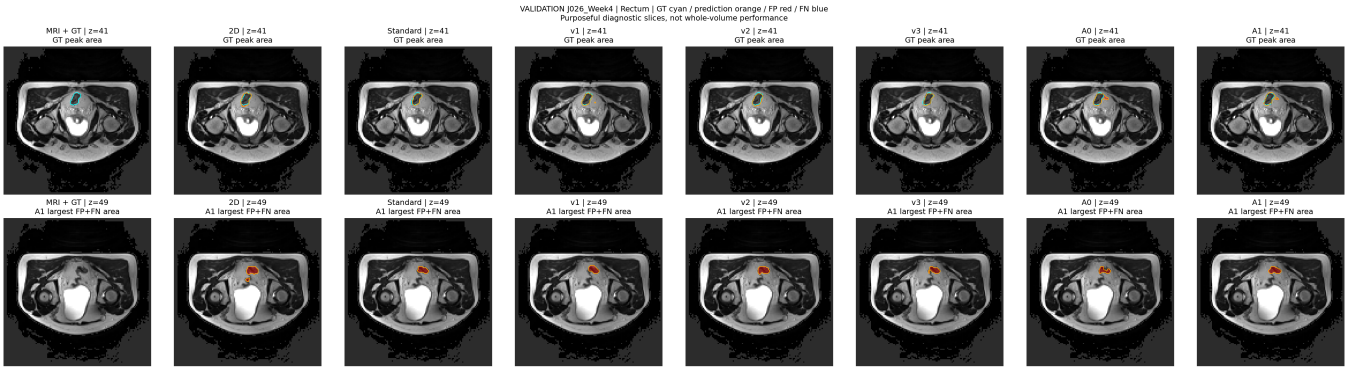
4.6.3 K018_Week4, Validation: Body and distant components



K018 Body errors

v3 greatly reduces the gross outer-volume pattern in v1, but residual distant predictions remain. Standard/v3 Body Dice is almost equal (0.9740/0.9742); HD95 is 4.75/6.24 mm, and distant Background-to-Body voxels are 5/3,128. A large correct interior dominates Dice while small distant components damage surface reliability. Normalisation/context is a supported association, not proof of one isolated cause.

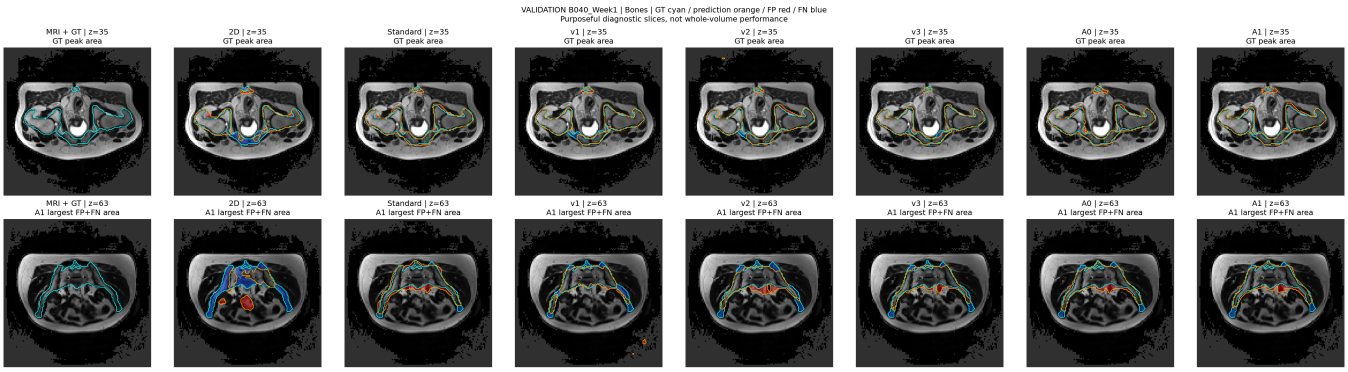
4.6.4 J026_Week4, Validation: rectum extent



J026 rectum errors

The difficult lower slice shows predictions beyond the labelled rectum; organ-centre agreement misses this extent failure. Whole-volume Dice improves 0.8110 to 0.8437, ASSD improves 2.00 to 1.73 mm, but HD95 remains 6.24 mm. Precision-recall and extent errors coexist with better average localisation; this is not a claim of anatomical correctness.

4.6.5 B040_Week1, Validation: bone coverage



B040 Bones errors

The pelvis contains both missing bone and extra predicted regions. v3 improves whole-volume Bones Dice 0.8607 to 0.9026 and HD95 7.41 to 4.58 mm, but the difficult lower slice retains FP and FN. Imbalance/coverage concerns are not limited to three small organs. Bones and Body must be reported explicitly rather than hidden by small-organ-only analysis.

5. Final discussion and engineering recommendation

5.1 RQ1: What did three-dimensional context contribute?

Relative to the supplementary 2D configuration, Standard and v3 improve whole-prostate Test Dice from 0.6611 to 0.7433/0.7485. Apex mean slice Dice increases from 0.5186 to 0.7544/0.7523, and base Dice from 0.3473 to 0.5097/0.5069. The patient-equal frequency of visits with internal predicted prostate gaps falls from 43.75% to 0.00%/8.33%. The area-transition measurements also more closely resemble GT. Together, these observations support the usefulness of the implemented 3D configurations for spatial consistency beyond easy central overlap.

This does not isolate a causal effect of dimensionality. The 2D run also differs in width, organ-aware sampling and class weights. The supported conclusion is that the tested 3D configurations performed better on these measurements, not that 3D kernels alone caused every gain. The low base-region means and the W029 example show that neighbouring-slice context still leaves ambiguous organ ends unresolved.

5.2 RQ2: Did Improved v3 solve the boundary problem beyond Standard?

The answer is mixed. v3 improves all five foreground Test mean Dice values, including a 0.0547 increase for Bladder and a smaller 0.0052 increase for Prostate. Median prostate HD95 improves by about 1.33 mm. These are useful average gains, and the bone and rectum examples show concrete benefits in some anatomy.

The evidence does not support consistent boundary superiority. Compared with Standard, v3 has slightly lower apex/base slice Dice, a higher patient-equal frequency of internal gaps, and more GT Bladder/Rectum voxels assigned to Prostate. Prostate precision falls from 0.7189 to 0.6983 while recall rises from 0.8415 to 0.8516. Worst prostate and rectum HD95 rise to 115.37 and 109.19 mm. K018 demonstrates how nearly equal Body Dice can coexist with thousands of distant false-positive voxels. Therefore, v3's average improvement does not establish more dependable local contours.

5.3 RQ3: Is the measured benefit worth the computation?

For the reference downsampled volume, Standard uses about 3.23 times the 2D peak allocated inference memory, and v3 about 4.54 times. v3 uses about 1.41 times Standard's memory and has 2.37 million versus 1.41 million parameters. Reference inference latencies are similar: 0.1452 s for v3 and 0.1552 s for Standard. Five repetitions on one case do not establish a general speed advantage.

The additional memory is feasible on the tested A100 and accompanies higher average segmentation scores. That supports using v3 as a research candidate when average organ segmentation is the objective. It does not establish value for unattended contouring, because local failures would still require review. These are inference measurements on downsampled volumes and small-width networks; they do not demonstrate a universal 5-10 times training-cost penalty or predict full-resolution deployment cost.

5.4 What mechanisms are supported, and what remains uncertain?

The earlier v1 had worse Body HD95 in all 38 Validation volumes: median 76.61 mm versus Standard 4.75 mm. Selected-case sliding-window inference with unchanged v1 weights reduced distant Body predictions but damaged small-organ overlap. Removing auxiliary outputs from a sum-trained model also severely degraded performance. Changing v1's InstanceNorm to GN1 in v3 reduced the gross Body failure and improved the original Validation score. These observations justify investigating normalisation, input context and output-head interactions.

They do not establish one universal cause. A main-only diagnostic changes how a sum-trained model is used; it is not a comparison with a separately retrained architecture. A numerical head-replay control failed with small logit differences, so it is not used as causal proof. A Train edge-sampling audit passed, but exposure checks alone cannot show a segmentation benefit. Loss weighting, limited patient diversity and optimisation variability remain plausible contributors. The tested A1/L/S configurations did not improve the overall result: these changed paired spatial augmentation, small-organ weights and the learning-rate schedule, respectively. The A1 comparison is quantified in Section 4.1; its result does not establish that augmentation is ineffective in other settings.

5.5 Generalisation and strength of evidence

v3 improves the original Validation primary score for five of six patients, but its existing Test small-organ gain occurs in only three of six. Excluding influential patient M015 changes the Test contrast with Standard from positive to approximately -0.0050. The same-recipe control C also differs from the earlier v3 run despite the same nominal seed and has higher Test prostate Dice. These observations limit claims of a stable, population-wide advantage.

All checkpoints were selected by Validation rather than by choosing the best Test epoch. Nevertheless, Test was repeatedly inspected during research, and no external cohort or independent multi-seed study was completed. Six patients per

evaluation split provide limited diversity. The results therefore document observed gains and regressions in this cohort; they do not estimate clinical reliability in a new hospital or patient population.

5.6 Conclusion and engineering recommendation

Retain v3 as the candidate selected by the original Validation rule and Standard as the implemented reference. The evidence supports improved average foreground segmentation, with some useful gains in bone coverage, rectum localisation and difficult bladder cases. It also shows persistent prostate-end errors, adjacent-organ confusion and distant components. A workflow using these outputs should retain human contour review, with explicit inspection of prostate ends and neighbouring organs; no automatic clinical acceptance threshold was established here.

If further research is commissioned, first reserve a new patient-independent holdout and estimate variability across seeds. Then test one controlled intervention using Train/Validation, with both average overlap and boundary/continuity outcomes specified in advance. The current evidence does not justify further tuning against the already inspected Test cohort. For this assessment, the strongest conclusion is that 3D context is useful in the tested configurations, while higher mean Dice and a more elaborate architecture are insufficient evidence of reliable clinical boundaries.

6. Artificial Intelligence Usage Disclosure

ChatGPT/Codex assisted with scaffolding/refactoring, debugging, audits, metric checks, literature interpretation, plotting and report drafting, including the standalone submission packaging. Verification included source/weight hashes, historical initialization and actual-weight CPU logit parity, paired geometry/metric tests, all-head gradients, recovery equivalence and recomputation from integer confusion counts. Historical GPU audits reproduced fixed predictions; the standalone migration replay also reproduced all 76 Validation matrices with matching input/weight/source identities. No new training or Test decoding was performed for that migration check. Failed controls and negative experiments were not converted into positive findings. The student remains personally responsible for every formula/code line, understanding the work and responding to reviewers; this disclosure is not independent certification of all AI output.

References and implementation attribution

1. Ronneberger, Fischer and Brox. *U-Net: Convolutional Networks for Biomedical Image Segmentation*. 2015. https://arxiv.org/abs/1505.04597

2. Cicek et al. *3D U-Net: Learning Dense Volumetric Segmentation from Sparse Annotation*. 2016. https://arxiv.org/abs/1606.06650

3. Isensee et al. *Brain Tumor Segmentation and Radiomics Survival Prediction: Contribution to the BRATS 2017 Challenge*. 2018. https://arxiv.org/abs/1802.10508

These motivate skips, volumetric context, residual blocks and multiscale supervision. GroupNorm, width, objective and preprocessing are explicit project adaptations; no paper is claimed to validate this exact HipMRI system. The package refactors this project's own verified implementation, not an externally pretrained model. Retain attribution when reusing components.

Appendix A. Environment and executable usage

A.1 Install and check

The tested local CPU environment is Python 3.11, PyTorch 2.13.0+cpu, NumPy 2.4.6, SciPy 1.17.1, nibabel 5.4.2 and Matplotlib 3.11.0. `requirements.txt` pins core packages. Historical GPU records specify Python 3.11.15, PyTorch 2.13.0+cu130 and NumPy 2.4.6 on A100; not all historical auxiliary versions were recorded. These CPU package versions are not a complete record of the historical GPU environment. Install the official CUDA PyTorch wheel compatible with your compute environment; numerical replay is separately verified.

```
cd recognition/HipMRI_UNet_s4906923
python -m pip install -r requirements.txt
python -B -m unittest discover -s tests -v
```

In the existing Windows Conda environment, `MKL_THREADING_LAYER=SEQUENTIAL` and `OMP_NUM_THREADS=1` avoided an OpenMP runtime conflict; do not use a duplicate-runtime override. Run GPU training on allocated compute nodes, not the login node. A typical fresh-run allocation is one A100, two CPU cores, up to six hours; load and IO vary.

A.2 Train, validate and save

```
python -B train.py train --model standard --dataset-root /path/to/HipMRI_Study_open --run-dir
/path/to/new_standard --device cuda
python -B train.py train --model v3 --dataset-root /path/to/HipMRI_Study_open --run-dir /path/to/new_v3 --device
 cuda
```

Training only decodes Train/Validation. Test filenames are enumerated for cohort validation but Test arrays are not loaded. Each completed batch atomically saves `latest.pt` with optimizer, RNG and position. Validation improvements save `best.pt`; epoch JSON and `training.png` are retained. `--keep-epoch-weights` additionally retains all epoch weights if space permits. Compact default storage differs from historical retention, not the mathematical recipe.

```
python -B train.py train --model v3 --dataset-root /path/to/HipMRI_Study_open --run-dir /path/to/new_v3 --device
 cuda --resume
```

Resume requires the same source/config/runtime identity and submission-format recovery files. It does not mutate historical research runs. Two writers are rejected by `.writer.lock`; after a force-kill, confirm no writer remains before removing that stale lock. New 50-epoch training is not required to evaluate existing selected weights.

A.3 Evaluate fixed weights and export an example

Supply trusted local checkpoints explicitly; weights are excluded from Git. The selected checkpoint hashes appear in `evidence/experiments.json`. Existing private backups can be used in place without copying into this folder. Historical files contain Python optimizer/RNG objects: load only trusted checkpoints.

```
python -B train.py evaluate --model v3 --dataset-root /path/to/HipMRI_Study_open --checkpoint
/path/to/v3/best.pt --split validation --output /path/to/new_validation.json --device cuda
python -B train.py evaluate --model v3 --dataset-root /path/to/HipMRI_Study_open --checkpoint
/path/to/v3/best.pt --split test --output /path/to/new_fixed_test.json --device cuda
python -B predict.py --model v3 --checkpoint /path/to/v3/best.pt --dataset-root /path/to/HipMRI_Study_open
--split validation --case K008_Week0 --output-dir /path/to/new_example --device cuda
```

Use `--model standard` with Standard weights for the baseline. Outputs: `metrics.json`, `overlay.png`, `prediction_downsampled.nii.gz`. Inference uses MRI only; paired GT is used afterwards for metrics/illustration. The exported NIfTI uses the downsampled affine, not original-resolution geometry. Outputs refuse overwrite. Without authorised MRI/weights, use the synthetic tests and inspect the recorded metrics; random weights must not be substituted for experimental results.

A.4 Completed Validation verification

Before submission, the standalone implementation reproduced all 76 Standard/v3 Validation confusion matrices, input and weight identities, and physical surface measurements. Job 682858 completed in 00:04:49 with exit 0:0 on 21 October 2026, using PyTorch 2.13.0+cu130. All seven package tests passed before that check. The Validation scores were 0.8502228594450512 (Standard) and 0.8563563984705311 (v3). Verification scripts, backup receipts and editorial checks are stored in the separate research archive; they are not needed to run the submitted models. This check establishes agreement with the earlier implementation, not new generalisation or clinical safety.

A.5 Export the report for Turnitin

```
python -m pip install reportlab==5.0.1
python -B export_report.py
```

`REPORT.pdf` is generated directly from this README and its figures. The feasibility review in Appendix C occupies one dedicated page. Review the export before submitting; this command does not upload to Turnitin.

Appendix B. Files, experiment records and assessment requirements

```
HipMRI_UNet_s4906923/
  modules.py             PyTorch-only Standard/v3 components
  dataset.py             Fixed patient split, paired preprocessing and sampling
  train.py               Direct model imports, training, Validation and fixed evaluation
  predict.py             Saved-model example, metrics, overlay and NIfTI export
  utils.py               Loss, inference, overlap and physical boundaries
  config/                Fixed patient split and training settings
  tests/                 Synthetic data, geometry, model and recovery tests
  evidence/              Ten-model count records, CSVs and source identities
  figures/               Method, curves, comparison and five real case panels
  README.md              This report, with a companion PDF export
```

The runtime has no import/path dependency on `20261017`, 2D or retired models. `evidence/experiments.json` preserves 500 epoch summaries and 680 best-model/case entries; the stored confusion counts allow readers to recalculate

overlap metrics. The archived research checks verified 400 foreground metric values, checkpoint selection and five original figure hashes. Source/checkpoint identities preserve research provenance without submitting MRI or parameter files. Packaging does not overwrite historical metrics.

The current eight-test suite covers split leakage rejection, NIfTI pairing, spatial resampling, label retention, padding/crop inversion, patient aggregation, anisotropic distances/empty masks, both models' gradients/loading, direct model construction in the trainer, and interrupted-versus-uninterrupted optimizer recovery. A synthetic 211-case file-list fixture ran the predict CLI end to end; it is not a research cohort or a performance result. Earlier tests also checked actual downloaded best-weight logits against historical implementations. The separate real-MRI GPU verification step has now also passed for all 76 Validation model/case comparisons; the runtime source hashes matched at the time of that check. Subsequent edits changed only comments and documentation in those files. Their executable syntax trees are unchanged; the comparison records are retained with the research archive.

Coursework requirement	Included evidence or remaining action
Hard volumetric residual model and deep supervision	Model code, architecture explanation, all-head gradients and six-class results
Implemented baseline, same split	Standard/v3, fixed split and input/weight identities
Per-class DSC/IoU and resource profiling	Six-class table, boundaries, parameters/VRAM/latency
Boundary/context dilemma and 3-5 failures	Five original MRI cases, prostate ends, OAR/slice diagnostics, recommendation
Modular code and reproducibility	Five required files, dependencies, commands and tests
Progressive originality	Existing staged Git history; adapted literature cited below
No datasets/model files in Git	Code, count records and selected report visuals only
Feasibility hurdle	Written review included; tutor registration must be confirmed
Upstream PR and code-review response	Lodge to upstream topic-recognition before the applicable deadline; respond during later feedback window
Turnitin	Submit PDF export of this README; confirm it matches the final committed report

Before submission, confirm the tutor check-off, course PR, Turnitin upload and applicable Blackboard deadline. The requirements above follow the local 21-page COMP3710 draft specification; check for later course announcements and follow the teaching team's code-review instructions.

Appendix C. One-page feasibility review

User need and scope. Evaluate a six-class improved 3D contouring prototype against an implemented baseline on supplied downsampled HipMRI. Prioritise patient independence, prostate precision and local boundary failures alongside overlap.

Acceptance criteria. (1) Paired grids/labels pass checks and patients never cross splits. (2) Implement volumetric residual context and multiscale supervision for the Hard route, reporting the stated Test Dice target of 0.7 with explicit aggregation. (3) Report all-class Dice/IoU, prostate precision/recall and physical boundary errors. (4) Measure peak allocated GPU memory, parameter count and reference latency. (5) Supply runnable commands, configuration, selection provenance and 3-5 representative failure analyses. These research criteria are not clinical safety thresholds.

Choice and course concepts. Standard provides volumetric encoder-decoder skip connections. v3 adds residual context, dropout and summed multiscale outputs. Normalisation uses one-group GroupNorm, written GroupNorm(1,C), where C is the number of channels. The work applies 3D convolution, multiclass supervised learning, imbalance weighting and patient-level generalisation analysis. It adapts the three cited papers rather than claiming full reproduction.

Feasibility evidence. There are 26/6/6 patients and 143/38/30 volumes. Ten formal models completed 50 epochs. Standard/v3 Validation small-organ scores are 0.8502/0.8564; Test scores are 0.7851/0.8145. The A100 profiling in Section 4.4 establishes operational feasibility. The original v3 training job completed in 02:02:48 including checkpoint IO and Validation; filesystem load affects wall time.

Risks, budget and fallback. Six evaluation patients per split, repeated visits, imbalance, input-context sensitivity and boundary outliers limit inference. Saving every epoch previously exhausted storage on the high-performance computing (HPC) system. The submission trainer saves latest/best by default, with optional epoch weights. Packaging and the standalone Validation check are now complete; confirm the course submission requirements before considering more tuning. Keep Standard as a fallback. Stronger claims about generalisation require evaluation on new, independent patients. The earlier proposed edge-sampling intervention was revised after diagnostics; its audit alone is not a completed segmentation improvement.

Consultation: this written review does not establish that the compulsory practical check-off was recorded. The student must confirm that separately with the tutor.